VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech
Faculty Name: Prof. S.Gopikrishnan
Student name: Raja Dhanish

Date: 26-02-2026
School: SCOPE
Reg. no.: 23BCE9845

---

1. Use MongoDB to implement the following DB operations

   1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".

```
mongosh mongodb://localho
Please enter a MongoDB connection string (Default: mongodb://localhost/): mongodb://localhost:27017
mongodb://localhost:27017
Current Mongosh Log ID: 69a06b7e4bee79dfb17c2906
Connecting to:          mongodb://localhost:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:          8.2.5
Using Mongosh:          2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-02-26T20:48:52.819+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> use vehicles
switched to db vehicles
```

   2. Write a MongoDB query to display all the databases.

```
mongosh mongodb://localho
vehicles> show dbs
admin     40.00 KiB
config    92.00 KiB
local     40.00 KiB
vehicles  80.00 KiB
vehicles> db.createCollection("two_wheelers", {
|    capped: true,
|    size: 500000,
|    max: 100
| })
{ ok: 1 }
vehicles> db.createCollection("four_wheelers")
{ ok: 1 }
vehicles> db.two_wheelers.insertMany([
| {
|   bike_name: "Tesla CyberBike",
|   model: "gearless",
|   category: "200cc",
|   colors_available: ["matte silver","neon cyan","midnight chrome"],
|   manufacturer: "Tesla",
|   performance: 9,
|   timestamp: new Date("2024-01-12"),
|   price: 350000
| },
| {
|   bike_name: "DragonFire X",
|   model: "gear",
|   category: "150cc",
|   colors_available: ["lava orange","carbon black","toxic green"],
|   manufacturer: "Inferno Motors",
|   performance: 8,
|   timestamp: new Date("2023-09-18"),
|   price: 165000
| },
| {
|   bike_name: "Phantom Glide",
|   model: "gearless",
|   category: "125cc",
|   colors_available: ["ghost white","smoke grey","electric purple"],
|   manufacturer: "Spectre Auto",
|   performance: 7,
```

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3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

```
vehicles> db.createCollection("two_wheelers", {
|     capped: true,
|     size: 500000,
|     max: 100
| })
{ ok: 1 }
vehicles> db.createCollection("four_wheelers")
{ ok: 1 }
```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
vehicles> db.two_wheelers.insertMany([
| {
|   bike_name: "Tesla CyberBike",
|   model: "gearless",
|   category: "200cc",
|   colors_available: ["matte silver","neon cyan","midnight chrome"],
|   manufacturer: "Tesla",
|   performance: 9,
|   timestamp: new Date("2024-01-12"),
|   price: 350000
| },
| {
|   bike_name: "DragonFire X",
|   model: "gear",
|   category: "150cc",
|   colors_available: ["lava orange","carbon black","toxic green"],
|   manufacturer: "Inferno Motors",
|   performance: 8,
|   timestamp: new Date("2023-09-18"),
|   price: 165000
| },
| {
|   bike_name: "Phantom Glide",
|   model: "gearless",
|   category: "125cc",
|   colors_available: ["ghost white","smoke grey","electric purple"],
|   manufacturer: "Spectre Auto",
|   performance: 7,
|   timestamp: new Date("2022-12-05"),
|   price: 98000
| },
| {
|   bike_name: "AeroBlade 300",
|   model: "gear",
|   category: "200cc",
|   colors_available: ["aqua blue","metallic gold","deep maroon"],
|   manufacturer: "Skyline Wheels",
|   performance: 9,
|   timestamp: new Date("2024-03-21"),
|   price: 210000
| },
```

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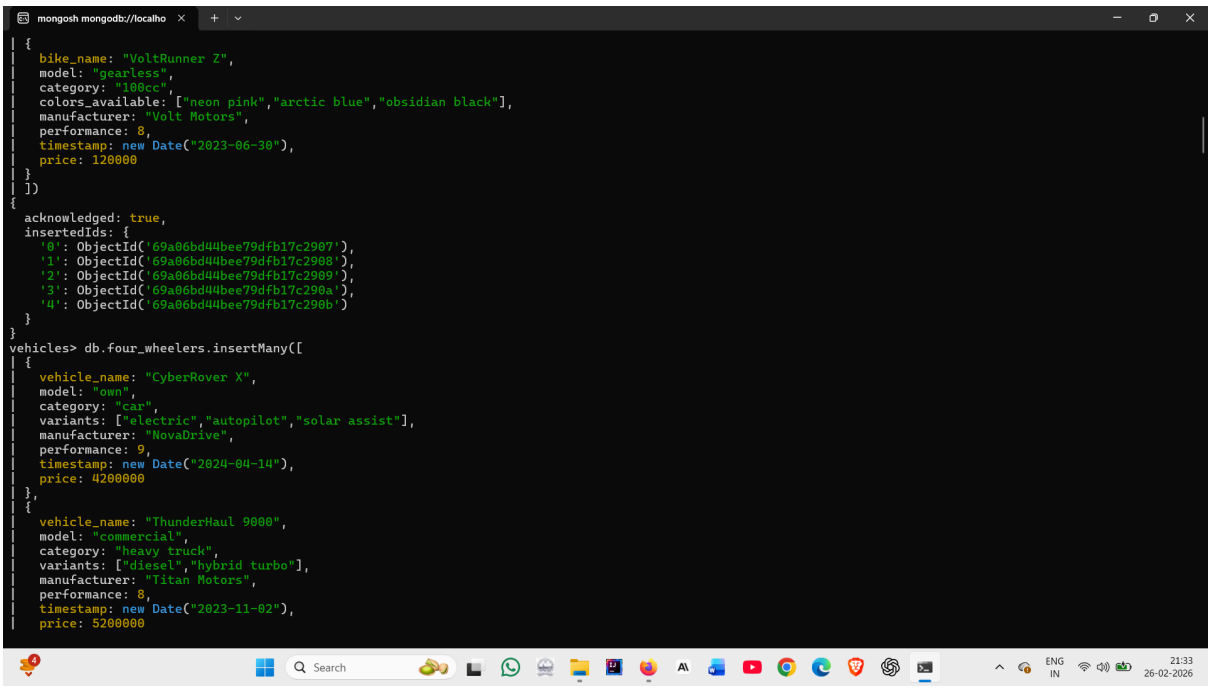

```
mongosh mongodb://localho
| {
|   bike_name: "VoltRunner Z",
|   model: "gearless",
|   category: "100cc",
|   colors_available: ["neon pink","arctic blue","obsidian black"],
|   manufacturer: "Volt Motors",
|   performance: 8,
|   timestamp: new Date("2023-06-30"),
|   price: 120000
| }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a06bd44bee79dfb17c2907'),
    '1': ObjectId('69a06bd44bee79dfb17c2908'),
    '2': ObjectId('69a06bd44bee79dfb17c2909'),
    '3': ObjectId('69a06bd44bee79dfb17c290a'),
    '4': ObjectId('69a06bd44bee79dfb17c290b')
  }
}
vehicles> db.four_wheelers.insertMany([
| {
|   vehicle_name: "CyberRover X",
|   model: "own",
|   category: "car",
|   variants: ["electric","autopilot","solar assist"],
|   manufacturer: "NovaDrive",
|   performance: 9,
|   timestamp: new Date("2024-04-14"),
|   price: 4200000
| },
| {
|   vehicle_name: "ThunderHaul 9000",
|   model: "commercial",
|   category: "heavy truck",
|   variants: ["diesel","hybrid turbo"],
|   manufacturer: "Titan Motors",
|   performance: 8,
|   timestamp: new Date("2023-11-02"),
|   price: 5200000
```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

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```
mongosh mongodb://localho
}
vehicles> db.four_wheelers.insertMany([
| {
|   vehicle_name: "CyberRover X",
|   model: "own",
|   category: "car",
|   variants: ["electric","autopilot","solar assist"],
|   manufacturer: "NovaDrive",
|   performance: 9,
|   timestamp: new Date("2024-04-14"),
|   price: 4200000
| },
| {
|   vehicle_name: "ThunderHaul 9000",
|   model: "commercial",
|   category: "heavy truck",
|   variants: ["diesel","hybrid turbo"],
|   manufacturer: "Titan Motors",
|   performance: 8,
|   timestamp: new Date("2023-11-02"),
|   price: 5200000
| },
| {
|   vehicle_name: "AquaCruiser",
|   model: "own",
|   category: "car",
|   variants: ["hydrogen","electric"],
|   manufacturer: "Hydra Mobility",
|   performance: 8,
|   timestamp: new Date("2025-01-10"),
|   price: 6100000
| },
| {
|   vehicle_name: "SkyShuttle Mini",
|   model: "commercial",
|   category: "mini truck",
|   variants: ["electric cargo","diesel"],
|   manufacturer: "Orbit Vehicles",
|   performance: 7,
|   timestamp: new Date("2023-06-25"),
|   price: 1800000
```

```
mongosh mongodb://localho
| {
|   vehicle_name: "AquaCruiser",
|   model: "own",
|   category: "car",
|   variants: ["hydrogen","electric"],
|   manufacturer: "Hydra Mobility",
|   performance: 8,
|   timestamp: new Date("2025-01-10"),
|   price: 6100000
| },
| {
|   vehicle_name: "SkyShuttle Mini",
|   model: "commercial",
|   category: "mini truck",
|   variants: ["electric cargo","diesel"],
|   manufacturer: "Orbit Vehicles",
|   performance: 7,
|   timestamp: new Date("2023-06-25"),
|   price: 1800000
| },
| {
|   vehicle_name: "UrbanBeast LX",
|   model: "own",
|   category: "SUV",
|   variants: ["petrol turbo","diesel AWD","hybrid"],
|   manufacturer: "Zenith Auto",
|   performance: 9,
|   timestamp: new Date("2024-08-19"),
|   price: 3900000
| }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a06c034bee79dfb17c290c'),
    '1': ObjectId('69a06c034bee79dfb17c290d'),
    '2': ObjectId('69a06c034bee79dfb17c290e'),
    '3': ObjectId('69a06c034bee79dfb17c290f'),
    '4': ObjectId('69a06c034bee79dfb17c2910')
  }
}
```

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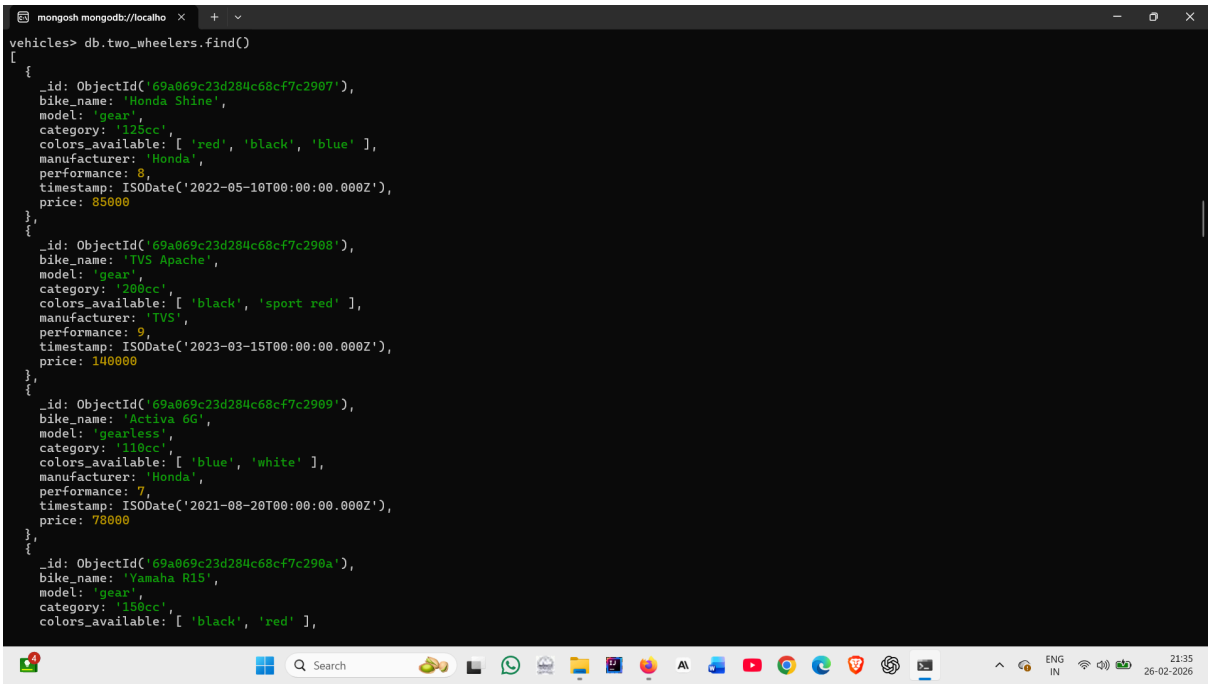
6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.



```
vehicles> db.two_wheelers.find()
[
  {
    _id: ObjectId('69a069c23d284c68cf7c2907'),
    bike_name: 'Honda Shine',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2022-05-10T00:00:00.000Z'),
    price: 85000
  },
  {
    _id: ObjectId('69a069c23d284c68cf7c2908'),
    bike_name: 'TVS Apache',
    model: 'gear',
    category: '200cc',
    colors_available: [ 'black', 'sport red' ],
    manufacturer: 'TVS',
    performance: 9,
    timestamp: ISODate('2023-03-15T00:00:00.000Z'),
    price: 140000
  },
  {
    _id: ObjectId('69a069c23d284c68cf7c2909'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '110cc',
    colors_available: [ 'blue', 'white' ],
    manufacturer: 'Honda',
    performance: 7,
    timestamp: ISODate('2021-08-20T00:00:00.000Z'),
    price: 78000
  },
  {
    _id: ObjectId('69a069c23d284c68cf7c290a'),
    bike_name: 'Yamaha R15',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'black', 'red' ],
```

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---

mongosh mongodb://localho

```
  },
  {
    _id: ObjectId('69a069c23d284c68cf7c290a'),
    bike_name: 'Yamaha R15',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'black', 'red' ],
    manufacturer: 'Yamaha',
    performance: 9,
    timestamp: ISODate('2022-11-01T00:00:00.000Z'),
    price: 180000
  },
  {
    _id: ObjectId('69a069c23d284c68cf7c290b'),
    bike_name: 'Bajaj Pulsar',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'black', 'blue' ],
    manufacturer: 'Bajaj',
    performance: 8,
    timestamp: ISODate('2020-06-18T00:00:00.000Z'),
    price: 120000
  },
  {
    _id: ObjectId('69a06bd44bee79dfb17c2907'),
    bike_name: 'Tesla CyberBike',
    model: 'gearless',
    category: '200cc',
    colors_available: [ 'matte silver', 'neon cyan', 'midnight chrome' ],
    manufacturer: 'Tesla',
    performance: 9,
    timestamp: ISODate('2024-01-12T00:00:00.000Z'),
    price: 350000
  },
  {
    _id: ObjectId('69a06bd44bee79dfb17c2908'),
    bike_name: 'DragonFire X',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'lava orange', 'carbon black', 'toxic green' ],
    manufacturer: 'Inferno Motors',
```

mongosh mongodb://localho

```
    category: '150cc',
    colors_available: [ 'lava orange', 'carbon black', 'toxic green' ],
    manufacturer: 'Inferno Motors',
    performance: 8,
    timestamp: ISODate('2023-09-18T00:00:00.000Z'),
    price: 165000
  },
  {
    _id: ObjectId('69a06bd44bee79dfb17c2909'),
    bike_name: 'Phantom Glide',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'ghost white', 'smoke grey', 'electric purple' ],
    manufacturer: 'Spectre Auto',
    performance: 7,
    timestamp: ISODate('2022-12-05T00:00:00.000Z'),
    price: 98000
  },
  {
    _id: ObjectId('69a06bd44bee79dfb17c290a'),
    bike_name: 'AeroBlade 300',
    model: 'gear',
    category: '200cc',
    colors_available: [ 'aqua blue', 'metallic gold', 'deep maroon' ],
    manufacturer: 'Skyline Wheels',
    performance: 9,
    timestamp: ISODate('2024-03-21T00:00:00.000Z'),
    price: 210000
  },
  {
    _id: ObjectId('69a06bd44bee79dfb17c290b'),
    bike_name: 'VoltRunner Z',
    model: 'gearless',
    category: '100cc',
    colors_available: [ 'neon pink', 'arctic blue', 'obsidian black' ],
    manufacturer: 'Volt Motors',
    performance: 8,
    timestamp: ISODate('2023-06-30T00:00:00.000Z'),
    price: 120000
  }
]
```

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```
mongosh mongodb://localho
vehicles> db.four_wheelers.find()
[
  {
    _id: ObjectId('69a06a643d284c68cf7c290c'),
    vehicle_name: 'Swift',
    model: 'own',
    category: 'car',
    variants: [ 'vxi', 'zxi', 'petrol' ],
    manufacturer: 'Maruti',
    performance: 7,
    timestamp: ISODate('2022-02-12T00:00:00.000Z'),
    price: 700000
  },
  {
    _id: ObjectId('69a06a643d284c68cf7c290d'),
    vehicle_name: 'Innova',
    model: 'own',
    category: 'car',
    variants: [ 'diesel', 'petrol' ],
    manufacturer: 'Toyota',
    performance: 9,
    timestamp: ISODate('2023-01-05T00:00:00.000Z'),
    price: 1900000
  },
  {
    _id: ObjectId('69a06a643d284c68cf7c290e'),
    vehicle_name: 'Tata Ace',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 6,
    timestamp: ISODate('2021-09-10T00:00:00.000Z'),
    price: 500000
  },
  {
    _id: ObjectId('69a06a643d284c68cf7c290f'),
    vehicle_name: 'Ashok Leyland Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
```

```
mongosh mongodb://localho
  {
    _id: ObjectId('69a06a643d284c68cf7c290f'),
    vehicle_name: 'Ashok Leyland Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 8,
    timestamp: ISODate('2020-04-22T00:00:00.000Z'),
    price: 2500000
  },
  {
    _id: ObjectId('69a06a643d284c68cf7c2910'),
    vehicle_name: 'Eicher Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
    performance: 7,
    timestamp: ISODate('2022-07-19T00:00:00.000Z'),
    price: 3000000
  },
  {
    _id: ObjectId('69a06c034bee79dfb17c290c'),
    vehicle_name: 'CyberRover X',
    model: 'own',
    category: 'car',
    variants: [ 'electric', 'autopilot', 'solar assist' ],
    manufacturer: 'NovaDrive',
    performance: 9,
    timestamp: ISODate('2024-04-14T00:00:00.000Z'),
    price: 4200000
  },
  {
    _id: ObjectId('69a06c034bee79dfb17c290d'),
    vehicle_name: 'ThunderHaul 9000',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel', 'hybrid turbo' ],
    manufacturer: 'Titan Motors',
    performance: 8,
```

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```
mongosh mongodb://localho
    category: 'heavy truck',
    variants: [ 'diesel', 'hybrid turbo' ],
    manufacturer: 'Titan Motors',
    performance: 8,
    timestamp: ISODate('2023-11-02T00:00:00.000Z'),
    price: 5200000
  },
  {
    _id: ObjectId('69a06c034bee79dfb17c290e'),
    vehicle_name: 'AquaCruiser',
    model: 'own',
    category: 'car',
    variants: [ 'hydrogen', 'electric' ],
    manufacturer: 'Hydra Mobility',
    performance: 8,
    timestamp: ISODate('2025-01-10T00:00:00.000Z'),
    price: 6100000
  },
  {
    _id: ObjectId('69a06c034bee79dfb17c290f'),
    vehicle_name: 'SkyShuttle Mini',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'electric cargo', 'diesel' ],
    manufacturer: 'Orbit Vehicles',
    performance: 7,
    timestamp: ISODate('2023-06-25T00:00:00.000Z'),
    price: 1800000
  },
  {
    _id: ObjectId('69a06c034bee79dfb17c2910'),
    vehicle_name: 'UrbanBeast LX',
    model: 'own',
    category: 'SUV',
    variants: [ 'petrol turbo', 'diesel AWD', 'hybrid' ],
    manufacturer: 'Zenith Auto',
    performance: 9,
    timestamp: ISODate('2024-08-19T00:00:00.000Z'),
    price: 3900000
  }
]
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

```
vehicles> db.two_wheelers.find({}, {bike_name:1, price:1, _id:0})
[
  { bike_name: 'Honda Shine', price: 85000 },
  { bike_name: 'TVS Apache', price: 140000 },
  { bike_name: 'Activa 6G', price: 78000 },
  { bike_name: 'Yamaha R15', price: 180000 },
  { bike_name: 'Bajaj Pulsar', price: 120000 },
  { bike_name: 'Tesla CyberBike', price: 350000 },
  { bike_name: 'DragonFire X', price: 165000 },
  { bike_name: 'Phantom Glide', price: 98000 },
  { bike_name: 'AeroBlade 300', price: 210000 },
  { bike_name: 'VoltRunner Z', price: 120000 }
]
```

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```
vehicles> db.four_wheelers.find({}, {vehicle_name:1, price:1, _id:0})
[
  { vehicle_name: 'Swift', price: 700000 },
  { vehicle_name: 'Innova', price: 1900000 },
  { vehicle_name: 'Tata Ace', price: 500000 },
  { vehicle_name: 'Ashok Leyland Truck', price: 2500000 },
  { vehicle_name: 'Eicher Bus', price: 3000000 },
  { vehicle_name: 'CyberRover X', price: 4200000 },
  { vehicle_name: 'ThunderHaul 9000', price: 5200000 },
  { vehicle_name: 'AquaCruiser', price: 6100000 },
  { vehicle_name: 'SkyShuttle Mini', price: 1800000 },
  { vehicle_name: 'UrbanBeast LX', price: 3900000 }
]
```

8. Write a MongoDB query to display two_wheelers from a particular company

```
vehicles> db.two_wheelers.find({manufacturer: "Honda"})
[
  {
    _id: ObjectId('69a069c23d284c68cf7c2907'),
    bike_name: 'Honda Shine',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2022-05-10T00:00:00.000Z'),
    price: 85000
  },
  {
    _id: ObjectId('69a069c23d284c68cf7c2909'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '110cc',
    colors_available: [ 'blue', 'white' ],
    manufacturer: 'Honda',
    performance: 7,
    timestamp: ISODate('2021-08-20T00:00:00.000Z'),
    price: 78000
  }
]
```

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9. Write a MongoDB query to display four_wheelers available in diesel variants

```
mongosh mongodb://localho
]
vehicles> db.four_wheelers.find({variants: "diesel"})
[
  {
    _id: ObjectId('69a06a643d284c68cf7c290d'),
    vehicle_name: 'Innova',
    model: 'own',
    category: 'car',
    variants: [ 'diesel', 'petrol' ],
    manufacturer: 'Toyota',
    performance: 9,
    timestamp: ISODate('2023-01-05T00:00:00.000Z'),
    price: 1900000
  },
  {
    _id: ObjectId('69a06a643d284c68cf7c290e'),
    vehicle_name: 'Tata Ace',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 6,
    timestamp: ISODate('2021-09-10T00:00:00.000Z'),
    price: 500000
  },
  {
    _id: ObjectId('69a06a643d284c68cf7c290f'),
    vehicle_name: 'Ashok Leyland Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 8,
    timestamp: ISODate('2020-04-22T00:00:00.000Z'),
    price: 2500000
  },
  {
    _id: ObjectId('69a06a643d284c68cf7c2910'),
    vehicle_name: 'Eicher Bus',
    model: 'commercial',
    category: 'bus',
```

```
mongosh mongodb://localho
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 8,
    timestamp: ISODate('2020-04-22T00:00:00.000Z'),
    price: 2500000
  },
  {
    _id: ObjectId('69a06a643d284c68cf7c2910'),
    vehicle_name: 'Eicher Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
    performance: 7,
    timestamp: ISODate('2022-07-19T00:00:00.000Z'),
    price: 3000000
  },
  {
    _id: ObjectId('69a06c034bee79dfb17c290d'),
    vehicle_name: 'ThunderHaul 9000',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel', 'hybrid turbo' ],
    manufacturer: 'Titan Motors',
    performance: 8,
    timestamp: ISODate('2023-11-02T00:00:00.000Z'),
    price: 5200000
  },
  {
    _id: ObjectId('69a06c034bee79dfb17c290f'),
    vehicle_name: 'SkyShuttle Mini',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'electric cargo', 'diesel' ],
    manufacturer: 'Orbit Vehicles',
    performance: 7,
    timestamp: ISODate('2023-06-25T00:00:00.000Z'),
    price: 1800000
  }
]
```

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Student name: Raja Dhanish

Date: 26-02-2026
School: SCOPE
Reg. no.: 23BCE9845

---

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```
mongosh mongodb://localho
]
vehicles> db.two_wheelers.find(
|   {performance: {$gt: 5}},
|   {bike_name:1, category:1, manufacturer:1, _id:0}
| )
|
[
  { bike_name: 'Honda Shine', category: '125cc', manufacturer: 'Honda' },
  { bike_name: 'TVS Apache', category: '200cc', manufacturer: 'TVS' },
  { bike_name: 'Activa 6G', category: '110cc', manufacturer: 'Honda' },
  { bike_name: 'Yamaha R15', category: '150cc', manufacturer: 'Yamaha' },
  { bike_name: 'Bajaj Pulsar', category: '150cc', manufacturer: 'Bajaj' },
  {
    bike_name: 'Tesla CyberBike',
    category: '200cc',
    manufacturer: 'Tesla'
  },
  {
    bike_name: 'DragonFire X',
    category: '150cc',
    manufacturer: 'Inferno Motors'
  },
  {
    bike_name: 'Phantom Glide',
    category: '125cc',
    manufacturer: 'Spectre Auto'
  },
  {
    bike_name: 'AeroBlade 300',
    category: '200cc',
    manufacturer: 'Skyline Wheels'
  },
  {
    bike_name: 'VoltRunner Z',
    category: '100cc',
    manufacturer: 'Volt Motors'
  }
]
vehicles> db.four_wheelers.find(
|   {performance: {$gt: 5}},
|   {vehicle_name:1, category:1, manufacturer:1, _id:0}
```

```
mongosh mongodb://localho
| )
[
  { vehicle_name: 'Swift', category: 'car', manufacturer: 'Maruti' },
  { vehicle_name: 'Innova', category: 'car', manufacturer: 'Toyota' },
  {
    vehicle_name: 'Tata Ace',
    category: 'mini truck',
    manufacturer: 'Tata'
  },
  {
    vehicle_name: 'Ashok Leyland Truck',
    category: 'heavy truck',
    manufacturer: 'Ashok Leyland'
  },
  { vehicle_name: 'Eicher Bus', category: 'bus', manufacturer: 'Eicher' },
  {
    vehicle_name: 'CyberRover X',
    category: 'car',
    manufacturer: 'NovaDrive'
  },
  {
    vehicle_name: 'ThunderHaul 9000',
    category: 'heavy truck',
    manufacturer: 'Titan Motors'
  },
  {
    vehicle_name: 'AquaCruiser',
    category: 'car',
    manufacturer: 'Hydra Mobility'
  },
  {
    vehicle_name: 'SkyShuttle Mini',
    category: 'mini truck',
    manufacturer: 'Orbit Vehicles'
  },
  {
    vehicle_name: 'UrbanBeast LX',
    category: 'SUV',
    manufacturer: 'Zenith Auto'
  }
]
```

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---

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.

```
mongosh mongodb://localho
Please enter a MongoDB connection string (Default: mongodb://localhost/): mongodb://localhost:27017
mongodb://localhost:27017
Current Mongosh Log ID: 69a071cdbcbb41dfe87c2906
Connecting to:          mongodb://localhost:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:          8.2.5
Using Mongosh:          2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-02-26T20:48:52.819+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> use animal
switched to db animal
animal> show dbs
admin      40.00 KiB
config     96.00 KiB
local      40.00 KiB
vehicles  144.00 KiB
```

2. Write a MongoDB query to display all the databases.

```
animal> show dbs
admin      40.00 KiB
config     96.00 KiB
local      40.00 KiB
vehicles  144.00 KiB
```

3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.

```
animal> db.createCollection("wild_animals", {
|     capped: true,
|     size: 500000,
|     max: 100
| })
{ ok: 1 }
animal> db.createCollection("domestic_animals")
{ ok: 1 }
```

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---

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
mongosh mongodb://localho
animal> db.wild_animals.insertMany([
| {
|   animal_name: "Bengal Tiger",
|   nature: "harm",
|   favorite_foods: ["deer", "wild boar"],
|   care_taker_name: "Dhanish",
|   life_span: 15,
|   timestamp: new Date(),
|   expenses: 50000
| },
| {
|   animal_name: "African Lion",
|   nature: "harm",
|   favorite_foods: ["zebra", "buffalo"],
|   care_taker_name: "Raja",
|   life_span: 14,
|   timestamp: new Date(),
|   expenses: 47000
| },
| {
|   animal_name: "King Cobra",
|   nature: "harm",
|   favorite_foods: ["rats", "frogs"],
|   care_taker_name: "Hemal",
|   life_span: 20,
|   timestamp: new Date(),
|   expenses: 15000
| },
| {
|   animal_name: "Sloth Bear",
|   nature: "harm",
|   favorite_foods: ["honey", "insects"],
|   care_taker_name: "Ravana",
|   life_span: 25,
|   timestamp: new Date(),
|   expenses: 30000
| },
| {
|   animal_name: "Indian Leopard",
|   nature: "harm",
|   favorite_foods: ["monkeys", "deer"],
```

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```
mongosh mongodb://localho
|   timestamp: new Date(),
|   expenses: 47000
| },
| {
|   animal_name: "King Cobra",
|   nature: "harm",
|   favorite_foods: ["rats", "frogs"],
|   care_taker_name: "Hemal",
|   life_span: 20,
|   timestamp: new Date(),
|   expenses: 15000
| },
| {
|   animal_name: "Sloth Bear",
|   nature: "harm",
|   favorite_foods: ["honey", "insects"],
|   care_taker_name: "Ravana",
|   life_span: 25,
|   timestamp: new Date(),
|   expenses: 30000
| },
| {
|   animal_name: "Indian Leopard",
|   nature: "harm",
|   favorite_foods: ["monkeys", "deer"],
|   care_taker_name: "Ramana",
|   life_span: 17,
|   timestamp: new Date(),
|   expenses: 42000
| }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a0721abcbb41dfe87c2907'),
    '1': ObjectId('69a0721abcbb41dfe87c2908'),
    '2': ObjectId('69a0721abcbb41dfe87c2909'),
    '3': ObjectId('69a0721abcbb41dfe87c290a'),
    '4': ObjectId('69a0721abcbb41dfe87c290b')
  }
}
```

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---

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

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Student name: Raja Dhanish

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```
mongosh mongodb://localho
animal> db.domestic_animals.insertMany([
| {
|   animal_name: "Persian Cat",
|   gender: "female",
|   favorite_foods: ["fish", "chicken"],
|   animal_petname: "Kitty",
|   life_span: 12,
|   timestamp: new Date(),
|   expenses: 8000
| },
| {
|   animal_name: "Labrador Dog",
|   gender: "male",
|   favorite_foods: ["meat", "biscuits"],
|   animal_petname: "Rocky",
|   life_span: 13,
|   timestamp: new Date(),
|   expenses: 10000
| },
| {
|   animal_name: "Jersey Cow",
|   gender: "female",
|   favorite_foods: ["grass", "fodder"],
|   animal_petname: "Gauri",
|   life_span: 18,
|   timestamp: new Date(),
|   expenses: 20000
| },
| {
|   animal_name: "White Rabbit",
|   gender: "male",
|   favorite_foods: ["carrot", "lettuce"],
|   animal_petname: "Snowy",
|   life_span: 8,
|   timestamp: new Date(),
|   expenses: 5000
| },
| {
|   animal_name: "Parrot Macaw",
|   gender: "female",
|   favorite_foods: ["seeds", "fruits"],
```

```
mongosh mongodb://localho
|   timestamp: new Date(),
|   expenses: 10000
| },
| {
|   animal_name: "Jersey Cow",
|   gender: "female",
|   favorite_foods: ["grass", "fodder"],
|   animal_petname: "Gauri",
|   life_span: 18,
|   timestamp: new Date(),
|   expenses: 20000
| },
| {
|   animal_name: "White Rabbit",
|   gender: "male",
|   favorite_foods: ["carrot", "lettuce"],
|   animal_petname: "Snowy",
|   life_span: 8,
|   timestamp: new Date(),
|   expenses: 5000
| },
| {
|   animal_name: "Parrot Macaw",
|   gender: "female",
|   favorite_foods: ["seeds", "fruits"],
|   animal_petname: "Chintu",
|   life_span: 10,
|   timestamp: new Date(),
|   expenses: 7000
| }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a07221bcbb41dfe87c290c'),
    '1': ObjectId('69a07221bcbb41dfe87c290d'),
    '2': ObjectId('69a07221bcbb41dfe87c290e'),
    '3': ObjectId('69a07221bcbb41dfe87c290f'),
    '4': ObjectId('69a07221bcbb41dfe87c2910')
  }
}
```

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---

6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.

```
mongosh mongodb://localho
animal> db.wild_animals.find()
[
  {
    _id: ObjectId('69a0721abcbb41dfe87c2907'),
    animal_name: 'Bengal Tiger',
    nature: 'harm',
    favorite_foods: [ 'deer', 'wild boar' ],
    care_taker_name: 'Dhanish',
    life_span: 15,
    timestamp: ISODate('2026-02-26T16:17:30.564Z'),
    expenses: 50000
  },
  {
    _id: ObjectId('69a0721abcbb41dfe87c2908'),
    animal_name: 'African Lion',
    nature: 'harm',
    favorite_foods: [ 'zebra', 'buffalo' ],
    care_taker_name: 'Raja',
    life_span: 14,
    timestamp: ISODate('2026-02-26T16:17:30.564Z'),
    expenses: 47000
  },
  {
    _id: ObjectId('69a0721abcbb41dfe87c2909'),
    animal_name: 'King Cobra',
    nature: 'harm',
    favorite_foods: [ 'rats', 'frogs' ],
    care_taker_name: 'Hemal',
    life_span: 20,
    timestamp: ISODate('2026-02-26T16:17:30.564Z'),
    expenses: 15000
  },
  {
    _id: ObjectId('69a0721abcbb41dfe87c290a'),
    animal_name: 'Sloth Bear',
    nature: 'harm',
    favorite_foods: [ 'honey', 'insects' ],
    care_taker_name: 'Ravana',
    life_span: 25,
    timestamp: ISODate('2026-02-26T16:17:30.564Z'),
    expenses: 30000
```

```
mongosh mongodb://localho
  },
  {
    _id: ObjectId('69a0721abcbb41dfe87c290b'),
    animal_name: 'Indian Leopard',
    nature: 'harm',
    favorite_foods: [ 'monkeys', 'deer' ],
    care_taker_name: 'Ramana',
    life_span: 17,
    timestamp: ISODate('2026-02-26T16:17:30.564Z'),
    expenses: 42000
  }
]
animal> db.domestic_animals.find()
[
  {
    _id: ObjectId('69a07221bcbb41dfe87c290c'),
    animal_name: 'Persian Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'chicken' ],
    animal_petname: 'Kitty',
    life_span: 12,
    timestamp: ISODate('2026-02-26T16:17:37.787Z'),
    expenses: 8000
  },
  {
    _id: ObjectId('69a07221bcbb41dfe87c290d'),
    animal_name: 'Labrador Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'biscuits' ],
    animal_petname: 'Rocky',
    life_span: 13,
    timestamp: ISODate('2026-02-26T16:17:37.787Z'),
    expenses: 10000
  },
  {
    _id: ObjectId('69a07221bcbb41dfe87c290e'),
    animal_name: 'Jersey Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'fodder' ],
    animal_petname: 'Gauri',
    life_span: 18,
```

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```
mongosh mongodb://localho
[
  {
    _id: ObjectId('69a07221bcbb41dfe87c290d'),
    animal_name: 'Labrador Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'biscuits' ],
    animal_petname: 'Rocky',
    life_span: 13,
    timestamp: ISODate('2026-02-26T16:17:37.787Z'),
    expenses: 10000
  },
  {
    _id: ObjectId('69a07221bcbb41dfe87c290e'),
    animal_name: 'Jersey Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'fodder' ],
    animal_petname: 'Gauri',
    life_span: 18,
    timestamp: ISODate('2026-02-26T16:17:37.787Z'),
    expenses: 20000
  },
  {
    _id: ObjectId('69a07221bcbb41dfe87c290f'),
    animal_name: 'White Rabbit',
    gender: 'male',
    favorite_foods: [ 'carrot', 'lettuce' ],
    animal_petname: 'Snowy',
    life_span: 8,
    timestamp: ISODate('2026-02-26T16:17:37.787Z'),
    expenses: 5000
  },
  {
    _id: ObjectId('69a07221bcbb41dfe87c2910'),
    animal_name: 'Parrot Macaw',
    gender: 'female',
    favorite_foods: [ 'seeds', 'fruits' ],
    animal_petname: 'Chintu',
    life_span: 10,
    timestamp: ISODate('2026-02-26T16:17:37.787Z'),
    expenses: 7000
  }
]
27°C
Mostly clear
Search
ENG
IN
21:54
26-02-2026
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

```
animal> db.wild_animals.find({}, {animal_name: 1, expenses: 1, _id: 0})
[
  { animal_name: 'Bengal Tiger', expenses: 50000 },
  { animal_name: 'African Lion', expenses: 47000 },
  { animal_name: 'King Cobra', expenses: 15000 },
  { animal_name: 'Sloth Bear', expenses: 30000 },
  { animal_name: 'Indian Leopard', expenses: 42000 }
]
animal> db.domestic_animals.find({}, {animal_name: 1, expenses: 1, _id: 0})
[
  { animal_name: 'Persian Cat', expenses: 8000 },
  { animal_name: 'Labrador Dog', expenses: 10000 },
  { animal_name: 'Jersey Cow', expenses: 20000 },
  { animal_name: 'White Rabbit', expenses: 5000 },
  { animal_name: 'Parrot Macaw', expenses: 7000 }
]
```

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---

8. Write a MongoDB query to display domestic_animals whose life is a particular year

```
animal> db.domestic_animals.find({life_span: 12})
[
  {
    _id: ObjectId('69a07221bcbb41dfe87c290c'),
    animal_name: 'Persian Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'chicken' ],
    animal_petname: 'Kitty',
    life_span: 12,
    timestamp: ISODate('2026-02-26T16:17:37.787Z'),
    expenses: 8000
  }
]
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker

```
animal> db.wild_animals.find({care_taker_name: "Raja"})
[
  {
    _id: ObjectId('69a0721abcbb41dfe87c2908'),
    animal_name: 'African Lion',
    nature: 'harm',
    favorite_foods: [ 'zebra', 'buffalo' ],
    care_taker_name: 'Raja',
    life_span: 14,
    timestamp: ISODate('2026-02-26T16:17:30.564Z'),
    expenses: 47000
  }
]
```

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School: SCOPE
Reg. no.: 23BCE9845

---

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
mongosh mongodb://localho
animal> db.wild_animals.find(
|   {life_span: {$gt: 5}},
|   {animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0}
| )
|
[
  {
    animal_name: 'Bengal Tiger',
    favorite_foods: [ 'deer', 'wild boar' ],
    expenses: 50000
  },
  {
    animal_name: 'African Lion',
    favorite_foods: [ 'zebra', 'buffalo' ],
    expenses: 47000
  },
  {
    animal_name: 'King Cobra',
    favorite_foods: [ 'rats', 'frogs' ],
    expenses: 15000
  },
  {
    animal_name: 'Sloth Bear',
    favorite_foods: [ 'honey', 'insects' ],
    expenses: 30000
  },
  {
    animal_name: 'Indian Leopard',
    favorite_foods: [ 'monkeys', 'deer' ],
    expenses: 42000
  }
]
animal> db.domestic_animals.find(
|   {life_span: {$gt: 5}},
|   {animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0}
| )
[
  {
    animal_name: 'Persian Cat',
    favorite_foods: [ 'fish', 'chicken' ],
    expenses: 8000
```

```
mongosh mongodb://localho
    favorite_foods: [ 'honey', 'insects' ],
    expenses: 30000
  },
  {
    animal_name: 'Indian Leopard',
    favorite_foods: [ 'monkeys', 'deer' ],
    expenses: 42000
  }
]
animal> db.domestic_animals.find(
|   {life_span: {$gt: 5}},
|   {animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0}
| )
[
  {
    animal_name: 'Persian Cat',
    favorite_foods: [ 'fish', 'chicken' ],
    expenses: 8000
  },
  {
    animal_name: 'Labrador Dog',
    favorite_foods: [ 'meat', 'biscuits' ],
    expenses: 10000
  },
  {
    animal_name: 'Jersey Cow',
    favorite_foods: [ 'grass', 'fodder' ],
    expenses: 20000
  },
  {
    animal_name: 'White Rabbit',
    favorite_foods: [ 'carrot', 'lettuce' ],
    expenses: 5000
  },
  {
    animal_name: 'Parrot Macaw',
    favorite_foods: [ 'seeds', 'fruits' ],
    expenses: 7000
  }
]
animal>
```